

YIXUAN LI

1088 Xueyuan Ave., Shenzhen, Guangdong, China
☎ +86 15920550976 ✉ shellyli1105@gmail.com 🌐 [ShellyLeee](#)

Profile

Studying Data Science as a senior at Southern University of Science and Technology (SUSTech), achieving a cumulative GPA of 3.79/4.00, ranking 8th out of 49, and maintaining an average score of 91.15/100.

Studied as a visiting student at University of California, San Diego (UCSD), achieving a GPA of 4.00/4.00.

Actively engaged in research on Uncertainty Estimation and Interpretability in large language models, with the goal of building more reliable and trustworthy AI systems.

Research Interests: Uncertainty Estimation, Interpretable AI, Hallucination Detection and Factuality Alignment, Calibrated Reward Modeling

Education

Southern University of Science and Technology

Major in Data Science

Sep. 2022 – Present

Shenzhen, China

- GPA: 3.79/4.00, Average Score: 91.15/100

University of California, San Diego

A visiting student at UCSD

Mar. 2025 – Jun. 2025

San Diego, CA, United States

- GPA: 4.00/4.00

Experience

Uncertainty Estimation in Large Language Models

Undergraduate Researcher

Feb. 2025 – Present

SUSTech, China

- Conducted a comprehensive literature review on LLM uncertainty estimation and developed a **standardized benchmarking pipeline** for comparative analysis.
- Evaluated **10+ semantic uncertainty methods** (e.g., SE, DSE, SAR) on QA datasets using open-source models, measuring performance with AUROC, AUARC, and PRR metrics.

Medical Data Analysis

Summer Internship

Jun. 2024 – Aug. 2024

Peking University Shenzhen Hospital, China

- Analyzed real-world clinical data from the department of dentistry, including dental records, X-ray data, and treatment history; performed de-duplication and standardization.
- Conducted exploratory data analysis and attempted feature selection for treatment prediction.

Projects

Semantic Uncertainty methods Evaluation Framework Construction | *Project Leader*

Jun. 2025 - Aug. 2025

- Engineered a unified pipeline to benchmark 10+ semantic uncertainty methods on some QA datasets with some open-source models.
- Performed a rigorous comparative study using uncertainty metrics to assess each method's ability to distinguish correct from incorrect model generations.

Interpretable Generative Model (CVPR 2025 Paper Reproduction) | *Core Member*

Apr. 2025 - Jun. 2025

- Reproduced the Post-hoc Concept Bottleneck (CB-AE and CC) pipeline based on the official implementation.
- Organized concept intervention experiments and replicated key results from the paper
- Designed an exploratory adversarial attack to evaluate model robustness under concept-level perturbations.

Fine-tuning Gemma-2B on Chinese QA Tasks (Kaggle Competition) | *Project Leader*

Oct. 2024 - Jan. 2025

- Implemented instruction tuning using LoRA on Gemma-2B with GPT-4 Chinese QA dataset.
- Built a complete fine-tuning pipeline: data preprocessing, prompt formatting, LoRA integration, and evaluation.
- Conducted multilingual evaluation to test model scalability; observed relatively robust performance on related languages.

Reinforcement Learning on FrozenLake (Course Project) | *Core Member*

Oct. 2024 - Jan. 2025

- Implemented algorithms (Double DQN, Dueling DQN), trained and analyzed agents on FrozenLake environment.
- Performed hyperparameter tuning and robustness comparison across algorithms, and achieve a performance with 98% accuracy.

Awards

• First-class scholarship	Awarded for outstanding students in SUSTech	2023 - 2024
• Third-class scholarship	Awarded for outstanding students in SUSTech	2022 - 2023
• Second Price	Awarded in the 2024 China Undergraduate Mathematical Contest in Modeling (CUMCM), in Guangdong division	Oct. 2024
• Honorable Mention	Awarded in the 2024 Mathematical Contest In Modeling (MCM)	May 2024

Skills

Programming Languages: Python, Java, JavaScript, SQL

Developer Tools: VS Code, IntelliJ IDEA, Google Cloud Platform

Software: Latex, Termius, Canva, Powerpoint, Jupyter Notebook, Git, OmniGraffle, Zotero, Xmind

Language: Mandarin (local), English (fluent: TOEFL 103), Cantonese (fluent)

Leadership / Extracurricular

Media of Shuren College

Sep. 2023 – Aug. 2024

President

SUSTech

- Leading a team with 15 Team members. Created a lots of content on college official account and organized activities.
- Award: Excellent College Student Leaders, Third Award of Excellent College Organization.

Pickleball Team

Oct. 2023 – Present

Member

SUSTech

- Being a Women’s doubles representing member of the school’s Pickleball team to compete in the second Guangdong Pickleball Competition, entering the top 16.